Chapter 23 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Are the p and q allele frequencies always equal when a population is in Hardy-Weinberg equilibrium? Explain your response.**(BIG IDEA 1)**

Concept 23.1: Genetic variation makes evolution possible

- **Genetic variation** refers to genetic differences among individuals within a population.
- The nucleotide differences that provide the basis of genetic variation originate when mutation and gene duplication produce new alleles and new genes. New genetic variants are produced rapidly in organisms with short generation times. In sexually reproducing organisms, most of the genetic differences among individuals result from crossing over, the independent assortment of chromosomes, and fertilization.

Typically, most of the nucleotide variability that occurs within a genetic locus does not affect the phenotype. Explain why.

Concept 23.2: The Hardy-Weinberg equation can be used to test whether a population is evolving

- A **population**, a localized group of organisms belonging to one species, is united by its **gene pool**, the aggregate of all the alleles in the population.
- For a population in **Hardy-Weinberg equilibrium**, the allele and genotype frequencies will remain constant if the population is large, mating is random, mutation is negligible, there is no gene flow, and there is no natural selection. For such a population, if p and q represent the frequencies of the only two possible alleles at a particular locus, then p2 is the frequency of one kind of homozygote, q2 is the frequency of the other kind of homozygote, and 2pq is the frequency of the heterozygous genotype.

Is it circular reasoning to calculate p *and* q *from observed genotype frequencies and then use those values of* p *and* q *to test if the population is in Hardy-Weinberg equilibrium? Explain your answer.*

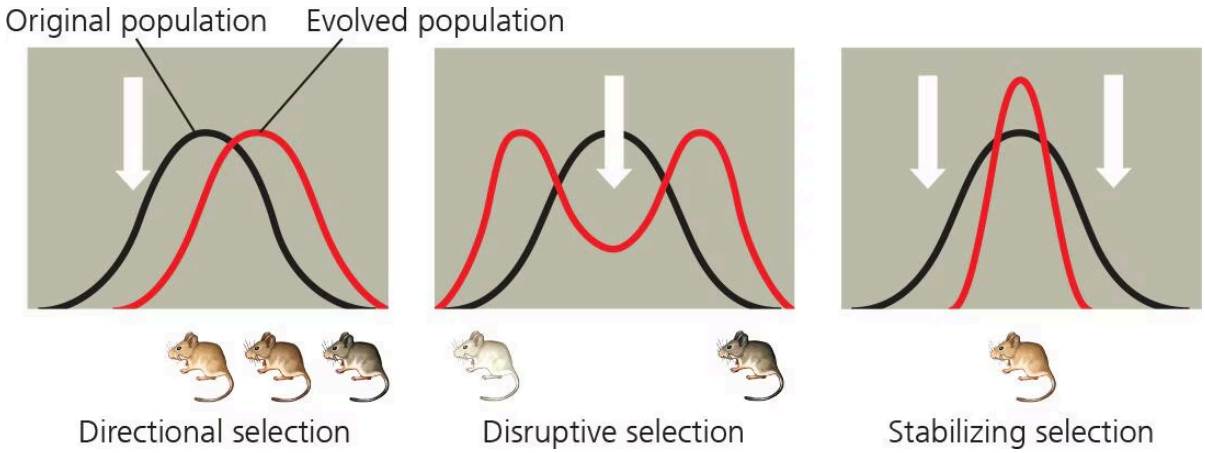
Concept 23.3: Natural selection, genetic drift, and gene flow can alter allele frequencies in a population

- In natural selection, individuals that have certain inherited traits tend to survive and reproduce at higher rates than other individuals *because of* those traits.
- In **genetic drift**, chance fluctuations in allele frequencies over generations tend to reduce genetic variation.
- ****Gene flow**, the transfer of alleles between populations, tends to reduce genetic differences between populations over time.

Would two small, geographically isolated populations in very different environments be likely to evolve in similar ways? Explain.

Concept 23.4: Natural selection is the only mechanism that consistently causes adaptive evolution

- One organism has greater **relative fitness** than another organism if it leaves more fertile descendants. The modes of natural selection differ in their effect on phenotype:



The original population is represented by a bell-shaped curve. In the graph for directional selection, the curve for the evolved population is bell-shaped, but it peaks to the right of the original population. An arrow points down to the left side of the original population. Below the graph are 3 mice from light brown to dark brown, with the second darkest mouse under the evolved population peak.

A second graph is given for disruptive selection. The evolved population curve rises, falls so it has a local minimum where the maximum for the original population is, then it rises and falls again. There is a white mouse to the left of the graphs and a dark mouse to the right.

A third graph is given for stabilizing selection. The evolved population curve is narrower and taller than the original population curve. A mouse is located directly below this peak; its coat is brown.

- Unlike genetic drift and gene flow, natural selection consistently increases the frequencies of alleles that enhance survival and reproduction, thus improving the degree to which organisms are well-suited for life in their environment.
- **Sexual selection** can result in secondary sex characteristics that can give individuals advantages in mating.
- **Balancing selection** occurs when natural selection maintains two or more forms in a population.
- There are constraints to evolution: Natural selection can act only on available variation; structures result from modified ancestral anatomy; adaptations are often compromises; and chance, natural selection, and the environment interact.

How might secondary sex characteristics in males differ from those in females in a species in which females compete for mates?